(within a millimeter of each other), on average, roughly every few seconds. But for a group of three molecules, you'd have to wait days, for four molecules you'd have to wait years, and for an initial dense blob of a million billion billion molecules it would take a length of time far greater than the current age of the universe for their random, dispersive motion to bring them back together into a small, ordered bunch. With more certainty than death and taxes, we can count on systems with many constituents evolving toward disorder.

Although it may not be immediately apparent, we have now come to an intriguing point. The second law of thermodynamics seems to have given us an arrow of time, *one that emerges when physical systems have a large number of constituents.* If you were to watch a film of a couple of carbon dioxide molecules that had been placed together in a small box (with a tracer showing the movements of each), you'd be hard pressed to say whether the film was running forward or in reverse. The two molecules would flit this way and that, sometimes coming together, sometimes moving apart, but they would not exhibit any gross, overall behavior distinguishing one direction in time from the reverse. However, if you were to watch a film of 10^{24} carbon dioxide molecules that had been placed together in the box (as a small, dense cloud of molecules, say), you could easily determine whether the film was being shown forward or in reverse: it is overwhelmingly likely that the forward time direction is the one in which the gas molecules become more and more uniformly spread out, *achieving higher and higher entropy.* If, instead, the film showed uniformly dispersed gas molecules swooshing together into a tight group, you'd immediately recognize that you were watching it in reverse.

The same reasoning holds for essentially all the things we encounter in daily life—things, that is, which have a large number of constituents: the forward-in-time arrow points in the direction of increasing entropy. If you watch a film of a glass of ice water placed on a bar, you can determine which direction is forward in time by checking that the ice melts—its H_2O molecules disperse throughout the glass, thereby achieving higher entropy. If you watch a film of a splattering egg, you can determine which direction is forward in time by checking that the